

ISEP INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

CSS



Sistemas de Informação 1

Content

- Context
 - Syntax
 - Integrating CSS into HTML
 - Selectors
 - By type of element
 - By identifier (#)
 - Por class (.)
 - By attributes
 - Based on combiners
 - By pseudo-classes
 - By pseudo-elements
 - Cascading
 - Inheritance
 - CSS grid
 - Best practices
-

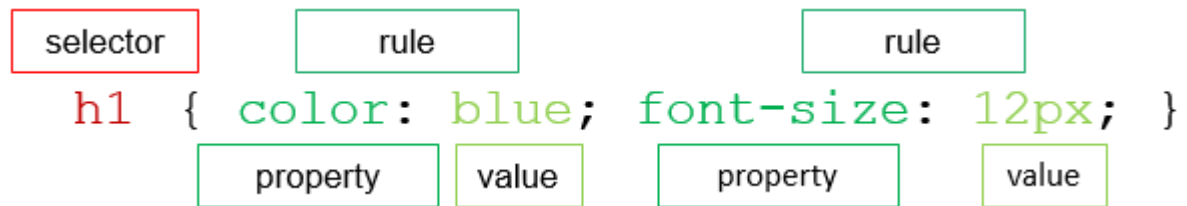
Context

CSS: Cascading Style Sheets

- Styling language
- Created and maintained by the W3C consortium
- We will be looking at version 3: <https://www.w3schools.com/css/>
- Used to add styling to a web page content
- Styles in a separate file
 - More readability, easier to maintain, allows code modularisation
- Advantages
 - Saves time: code can be reused in other web pages
 - Fast loading: does not need HTML attributes for each tag
 - Easier to maintain: the web page style can be altered by changing only the CSS file
 - Platform independent: CSS is a standard and supports all browsers

Syntax

The styles are defined as a set of rules



Components

- **Selector:** selects the HTML elements to style
- **Set of rules:**
 - One or more style rules surrounded by curly braces ({ and }).
 - Each rule has a property: value pair and ends with a semicolon (;).
 - Comments are made with /* and */

Integrating CSS into HTML

■ Local



```
<h1 style="color:blue;margin-left:30px;">I am the header</h1>
```

■ External



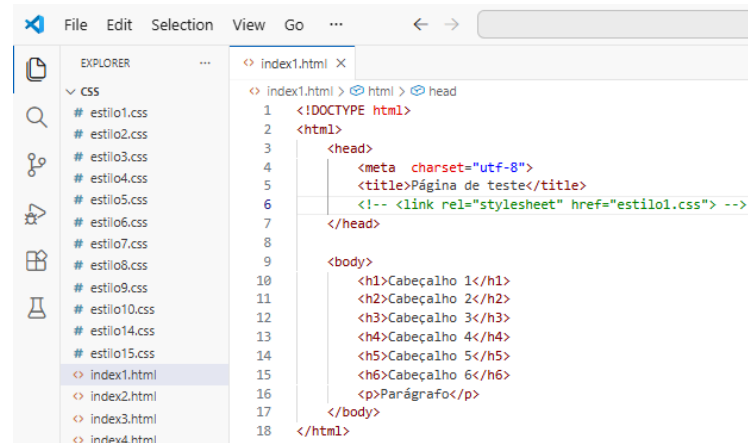
```
<head>  
  <link rel="stylesheet" type="text/css" href="style.css">  
</head>
```

■ Internal



```
<html>  
  <head>  
    <style>  
      body {background-color: red;}  
    ...  
  </style>  
</head>  
<body>  
  <h1> I am the header </h1>  
</body>  
</html>
```

Selector by type of element

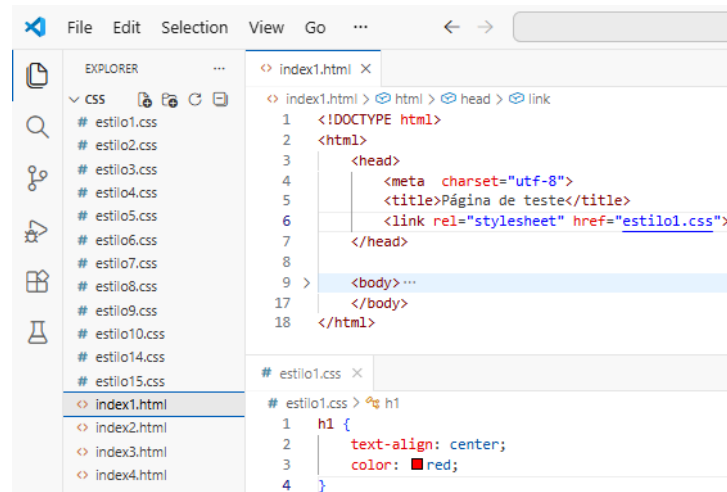


VS Code Explorer and Editor showing the index1.html file structure. The Explorer pane on the left shows a list of files: css (estilo1.css to estilo15.css), index1.html, index2.html, index3.html, and index4.html. The Editor pane shows the content of index1.html, which includes a head section with meta, title, and link tags, and a body section with h1 through h6 and a paragraph.

```

1 <!DOCTYPE html>
2 <html>
3   <head>
4     <meta charset="utf-8">
5     <title>Página de teste</title>
6     <!-- <link rel="stylesheet" href="estilo1.css" -->
7   </head>
8
9   <body>
10    <h1>Cabeçalho 1</h1>
11    <h2>Cabeçalho 2</h2>
12    <h3>Cabeçalho 3</h3>
13    <h4>Cabeçalho 4</h4>
14    <h5>Cabeçalho 5</h5>
15    <h6>Cabeçalho 6</h6>
16    <p>Parágrafo</p>
17  </body>
18 </html>

```



VS Code Explorer and Editor showing the index1.html file structure. The Explorer pane on the left shows a list of files: css (estilo1.css to estilo15.css), index1.html, index2.html, index3.html, and index4.html. The Editor pane shows the content of index1.html, which includes a head section with meta, title, and link tags, and a body section with h1 through h6 and a paragraph. The CSS rule for h1 is selected in the Editor pane.

```

1 <!DOCTYPE html>
2 <html>
3   <head>
4     <meta charset="utf-8">
5     <title>Página de teste</title>
6     <link rel="stylesheet" href="estilo1.css">
7   </head>
8
9   <body> ...
17 </body>
18 </html>

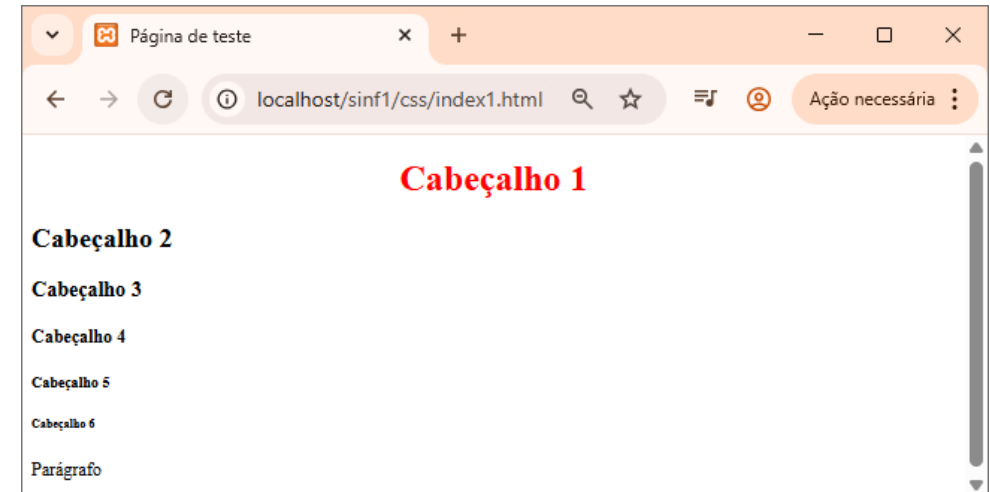
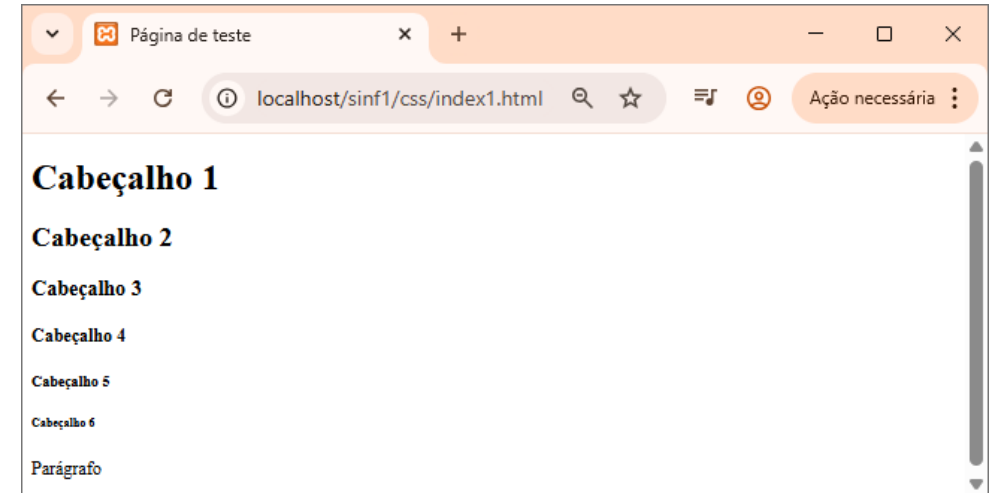
```

estilo1.css

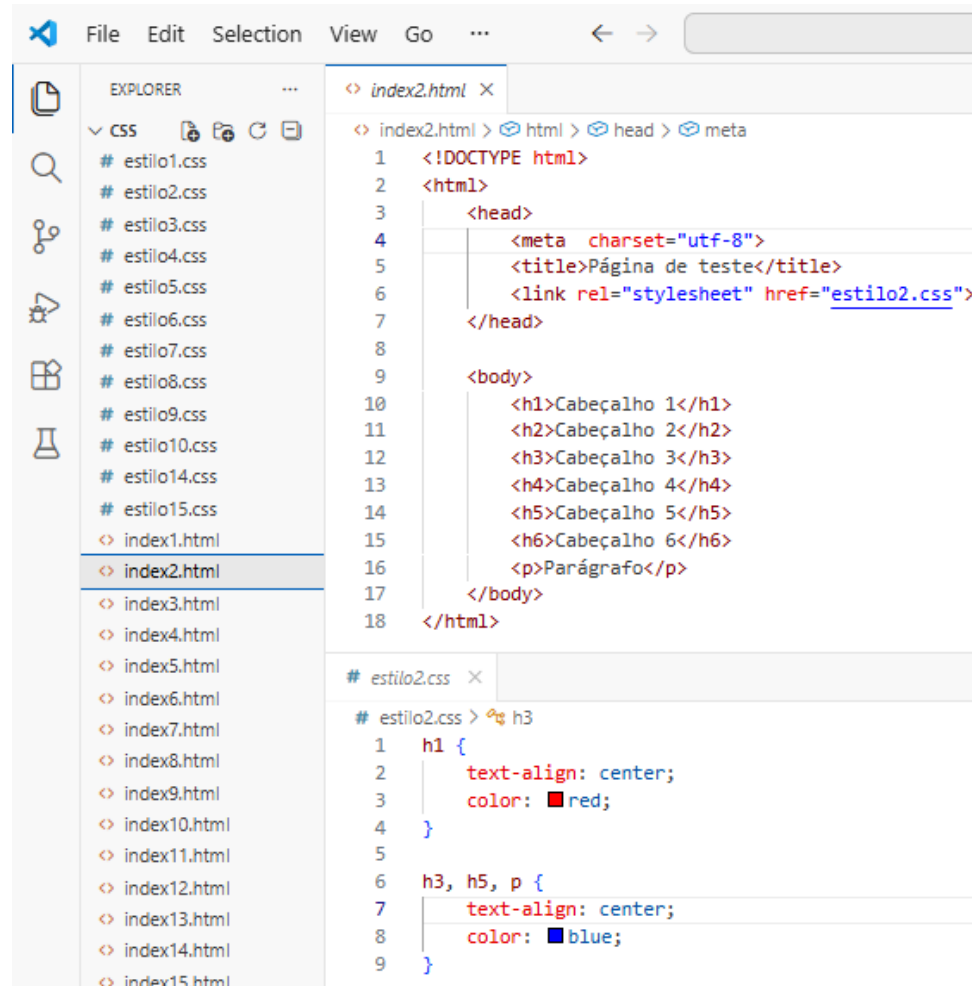
```

1 h1 {
2   text-align: center;
3   color: red;
4 }

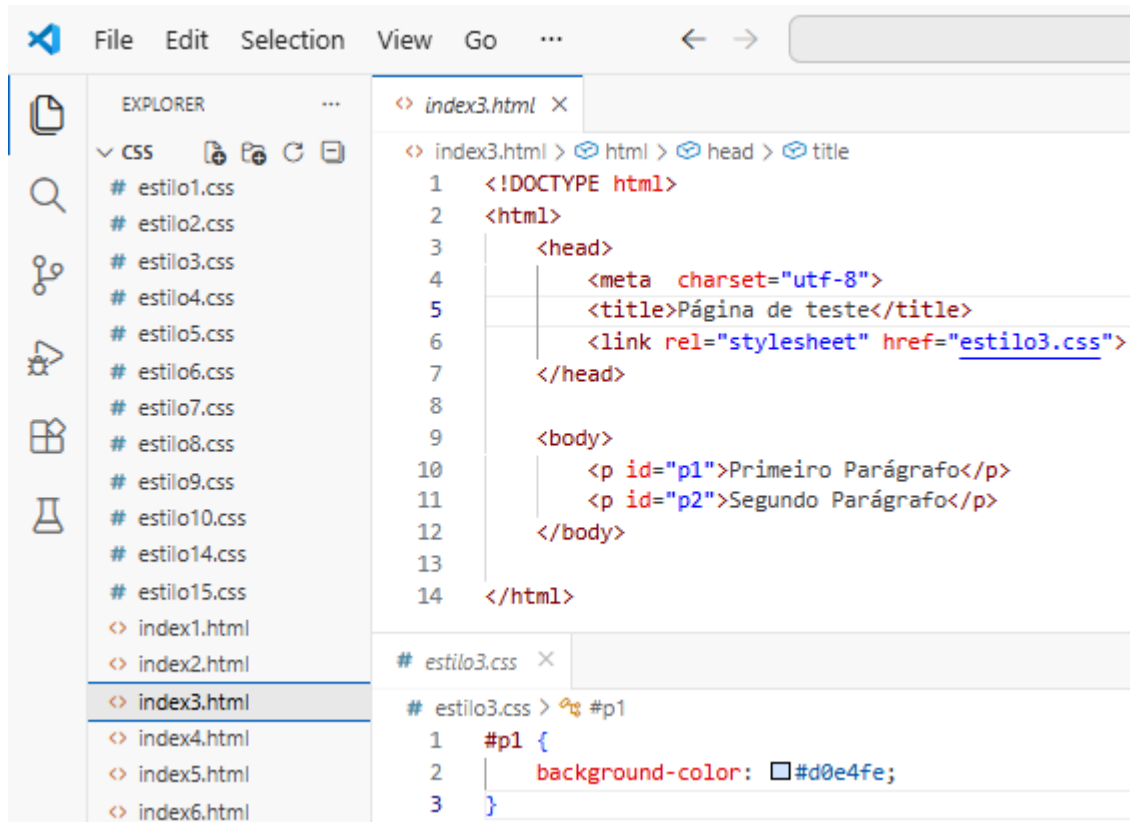
```



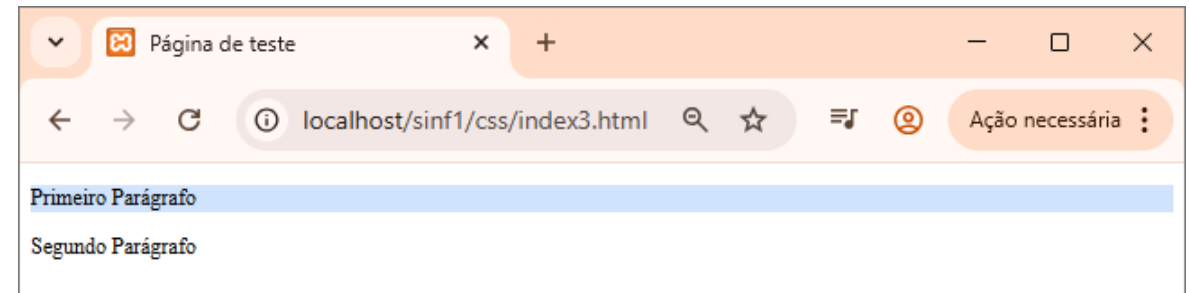
Selector by several types of elements



Selector by identifier (#)

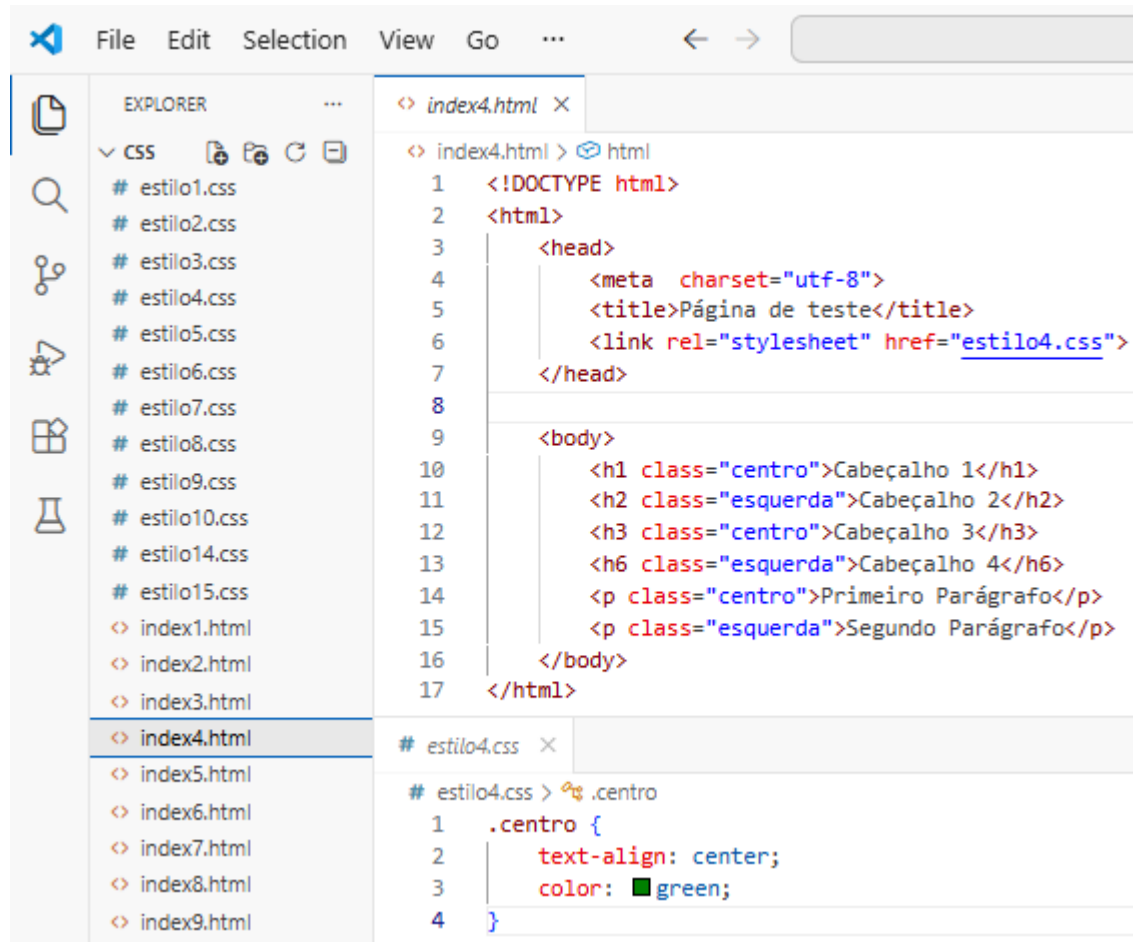


Selects HTML elements based on their `id`
before the `id`

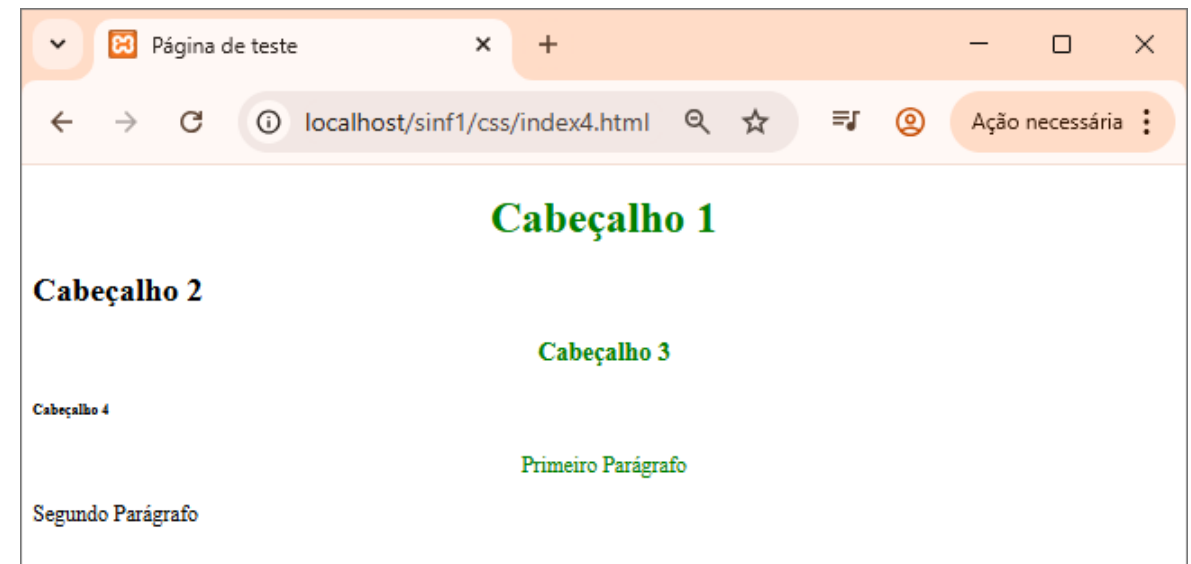


Choose hexadecimal colours:
https://www.w3schools.com/colors/colors_picker.asp

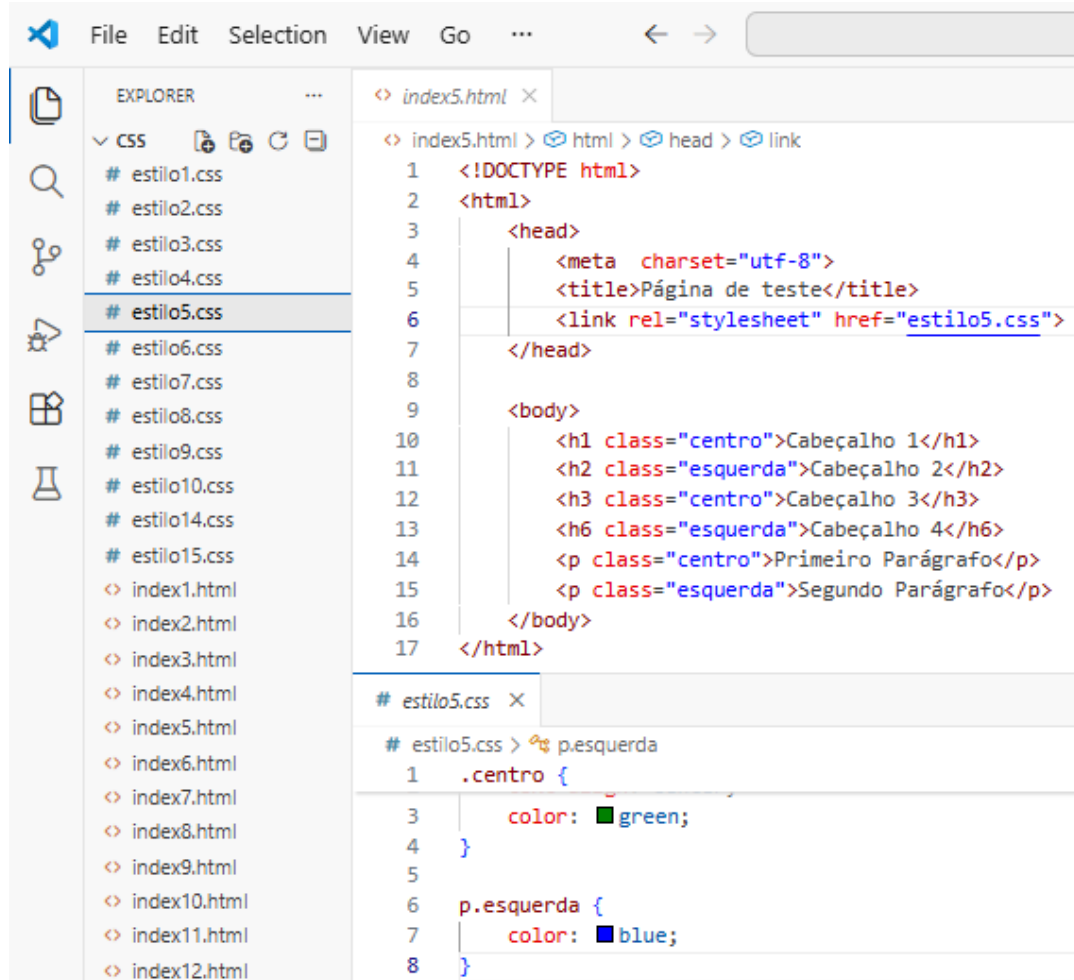
Selectors by class (.)



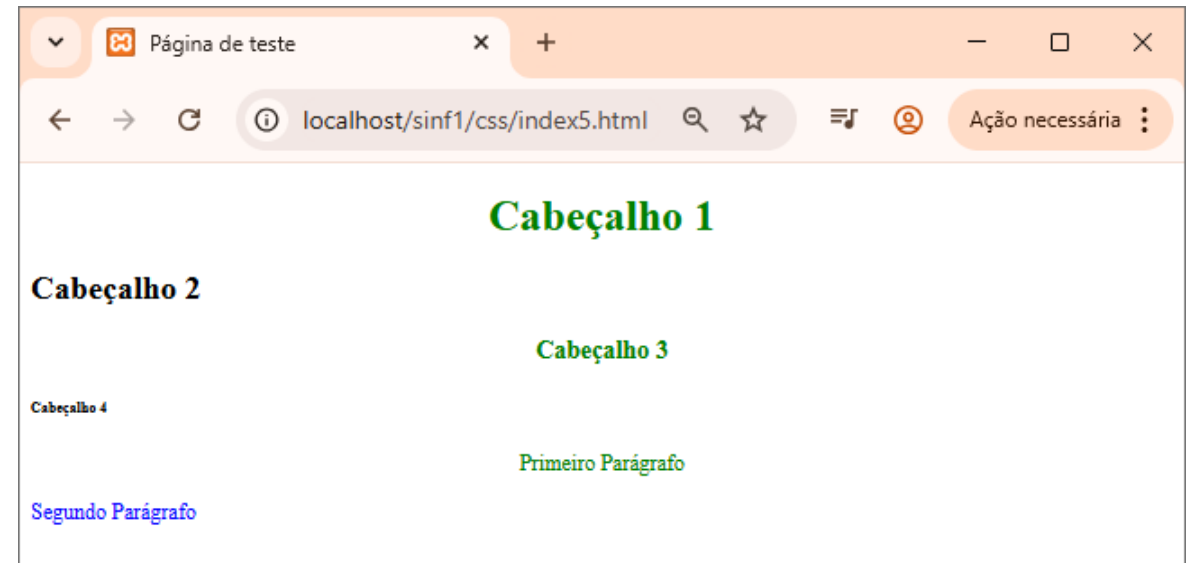
Selects HTML elements based on their `class`
. Before the `class`



Selectors by class (.)



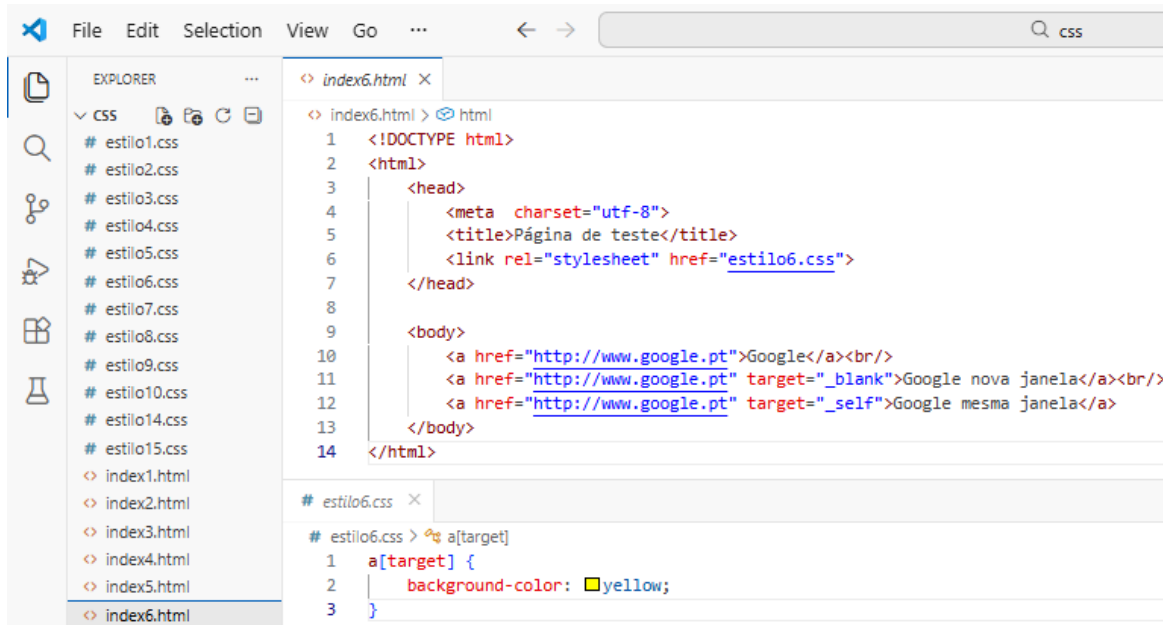
Selects specific HTML elements based on their `class`
.
Between the `tag` and the `class`



Selectors by attributes

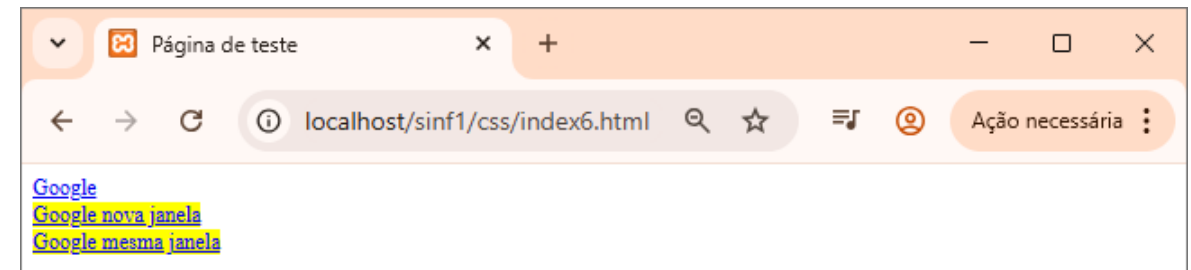
Selector	Description
<code>[attribute~="value"]</code>	Attribute value <u>contains</u> "value" (<u>full word</u>)
<code>[attribute ="value"]</code>	Attribute value <u>starts with</u> "value" (<u>full word</u>)
<code>[attribute^="value"]</code>	Attribute value <u>starts with</u> "value"
<code>[attribute\$="value"]</code>	Attribute value <u>ends with</u> "value"
<code>[attribute*="value"]</code>	Attribute value <u>contains</u> "value"

Selectors by attributes

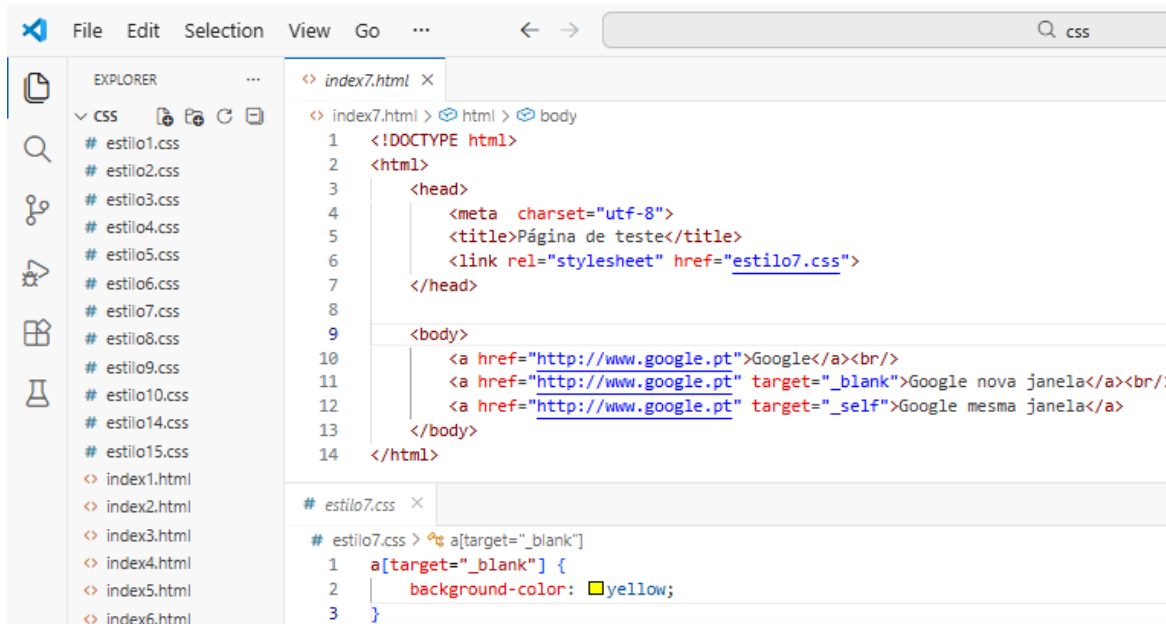


```
File Edit Selection View Go ...  
EXPLORER  
CSS  
# estilo1.css  
# estilo2.css  
# estilo3.css  
# estilo4.css  
# estilo5.css  
# estilo6.css  
# estilo7.css  
# estilo8.css  
# estilo9.css  
# estilo10.css  
# estilo14.css  
# estilo15.css  
index1.html  
index2.html  
index3.html  
index4.html  
index5.html  
index6.html  
index6.html  
estilo6.css  
# estilo6.css > a[target]  
1 a[target] {  
2   background-color: yellow;  
3 }
```

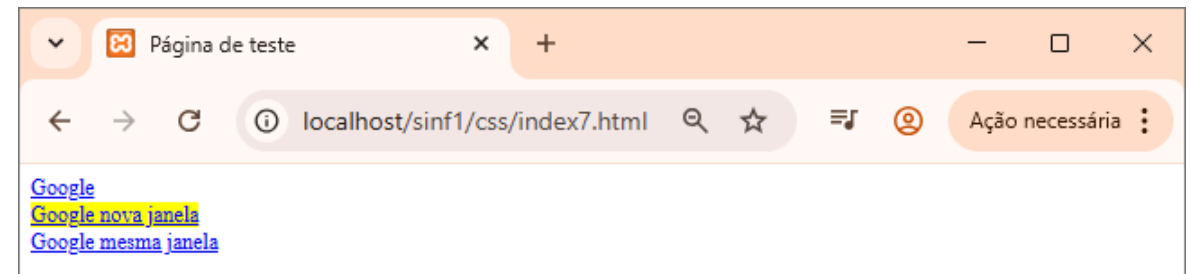
Specific attributes: `selector[attribute]` to select elements with a specific attribute.



Selectors by attributes



Attribute values: `selector[attribute=value]` select elements with specific attributes and values



Selectors based on combinators

EXPLORER

- css
 - # estilo1.css
 - # estilo2.css
 - # estilo3.css
 - # estilo4.css
 - # estilo5.css
 - # estilo6.css
 - # estilo7.css
 - # estilo8.css
 - # estilo9.css
 - # estilo10.css
 - # estilo14.css
 - # estilo15.css
- index.html
- index2.html
- index3.html
- index4.html
- index5.html
- index6.html
- index7.html
- index8.html
- index9.html
- index10.html
- index11.html

index8.html

```

1 <!DOCTYPE html>
2 <html>
3   <head>
4     <meta charset="utf-8">
5     <title>Página de teste</title>
6     <link rel="stylesheet" href="estilo8.css">
7   </head>
8
9   <body>
10    <div>
11      <p>Primeiro parágrafo</p>
12      <p>Segundo parágrafo</p>
13    </div>
14    <span>
15      <p>Terceiro parágrafo</p>
16      <p>Quarto parágrafo</p>
17    </span>
18    <p>Quinto parágrafo</p>
19  </body>
20 </html>

```

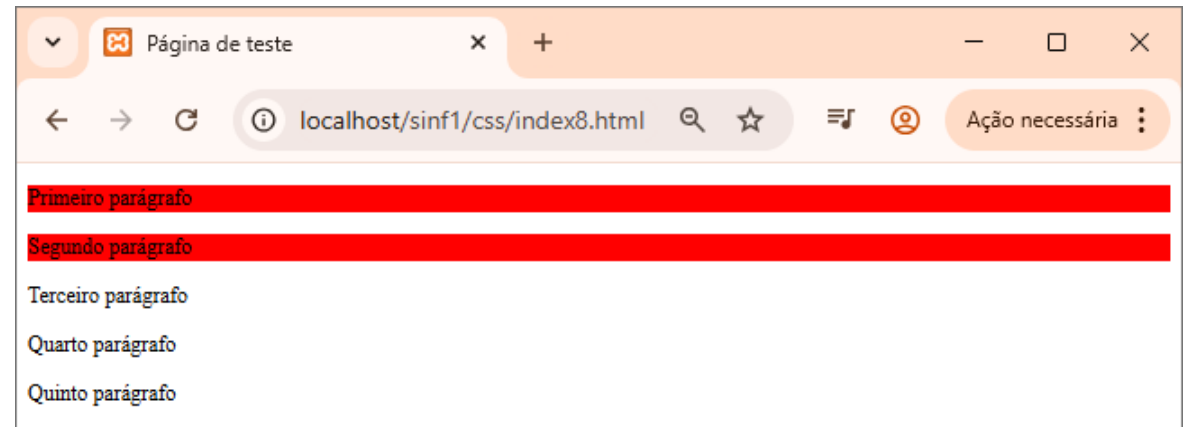
estilo8.css

```

1 div > p {
2   background-color: red;
3 }

```

Selectors	Description
Children ()	All the element's children and their children
Right child (>)	The elements children
Right sibling (+)	The siblings after the element
Sibling (~)	All the siblings



Selectors by pseudo-classes & by pseudo-elements

- **Pseudo-classes** define element status (e.g.: mouse-over, visited/not visited links, elements based on their location)

Syntax:

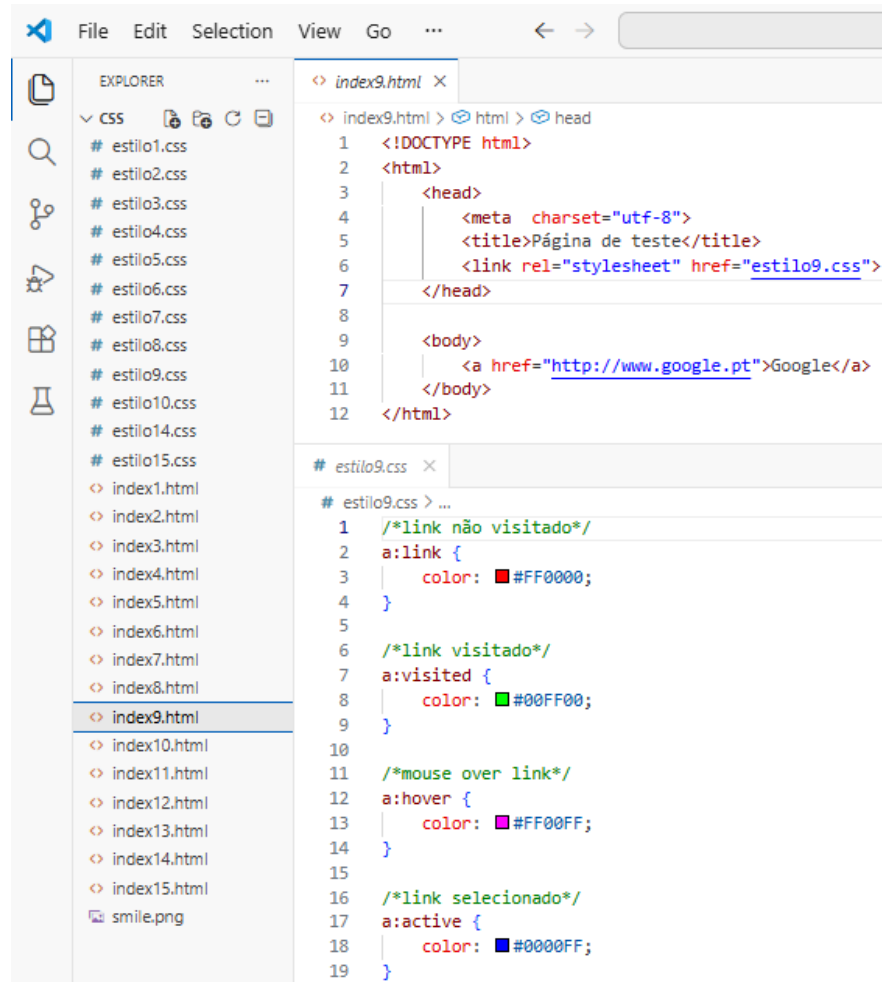
```
selector: pseudoclass{  
    property: value;  
}
```

- **Pseudo-elements** are used to style parts of an element (e.g: element's first row or character, insert content before/after the element)

Syntax:

```
selector:pseudoelement{  
    property: value;  
}
```

Selectors by pseudo-classes



The screenshot shows a code editor with two files open. The left pane shows the Explorer view with a list of files: CSS files (estilo1.css to estilo15.css) and HTML files (index1.html to index15.html). The right pane shows the content of index9.html and estilo9.css.

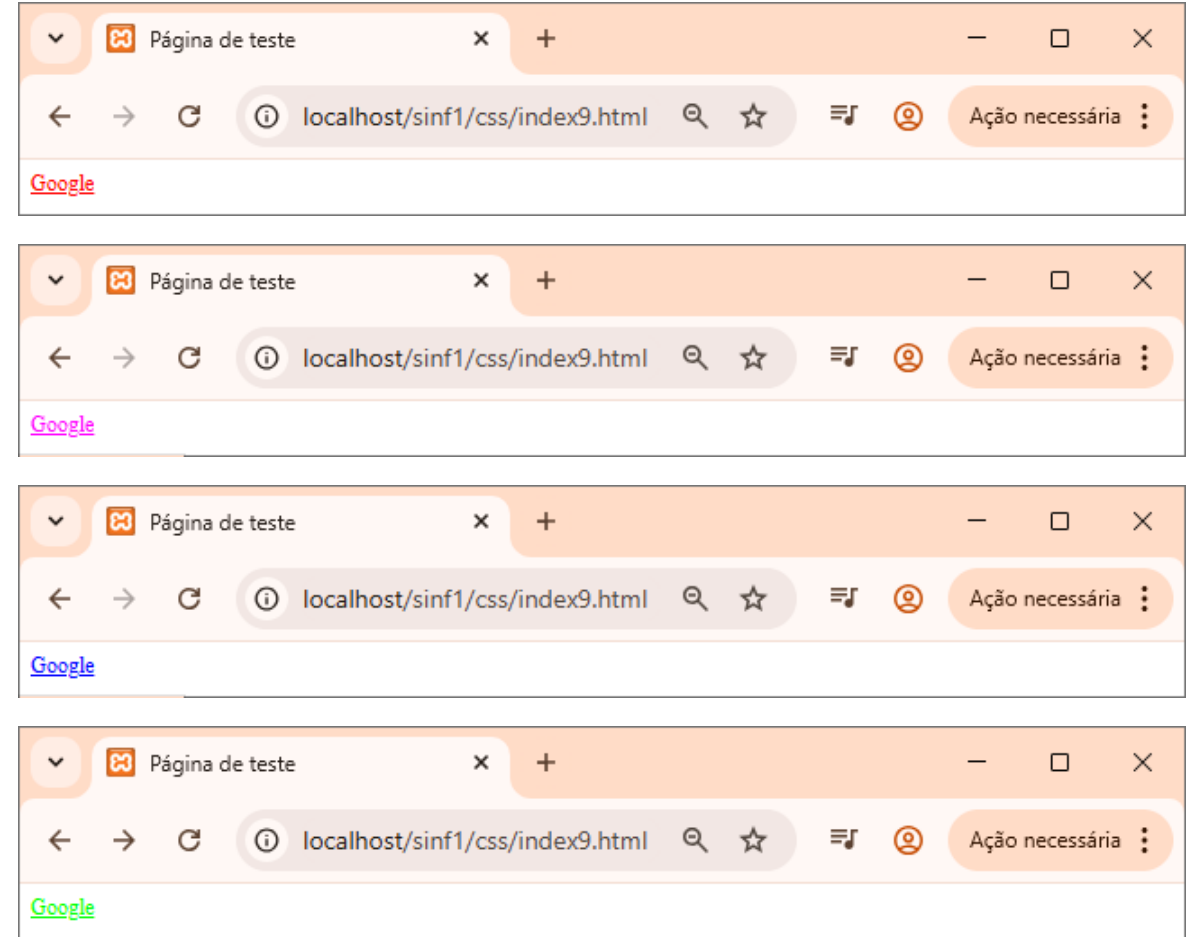
```
<?DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Página de teste</title>
    <link rel="stylesheet" href="estilo9.css">
  </head>
  <body>
    <a href="http://www.google.pt">Google</a>
  </body>
</html>
```

```
/*link não visitado*/
a:link {
  color: #FF0000;
}

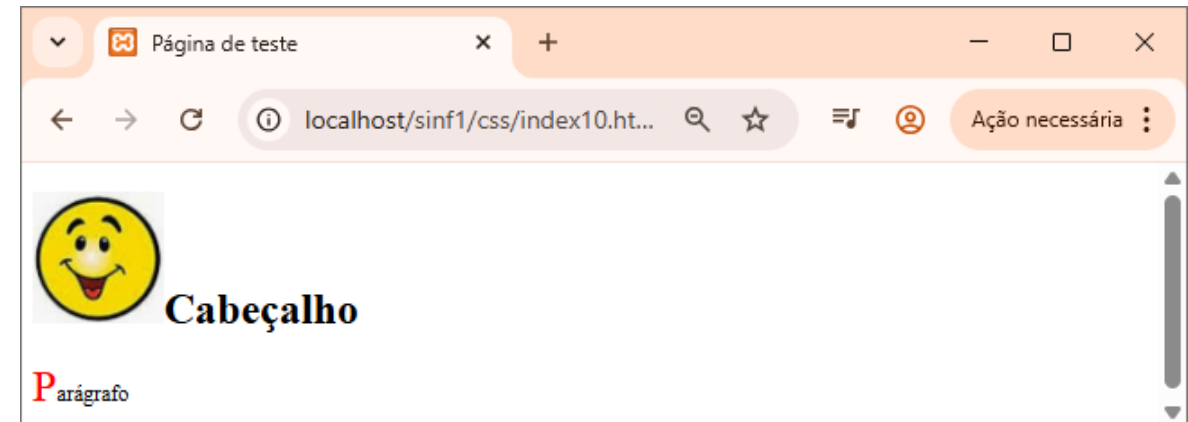
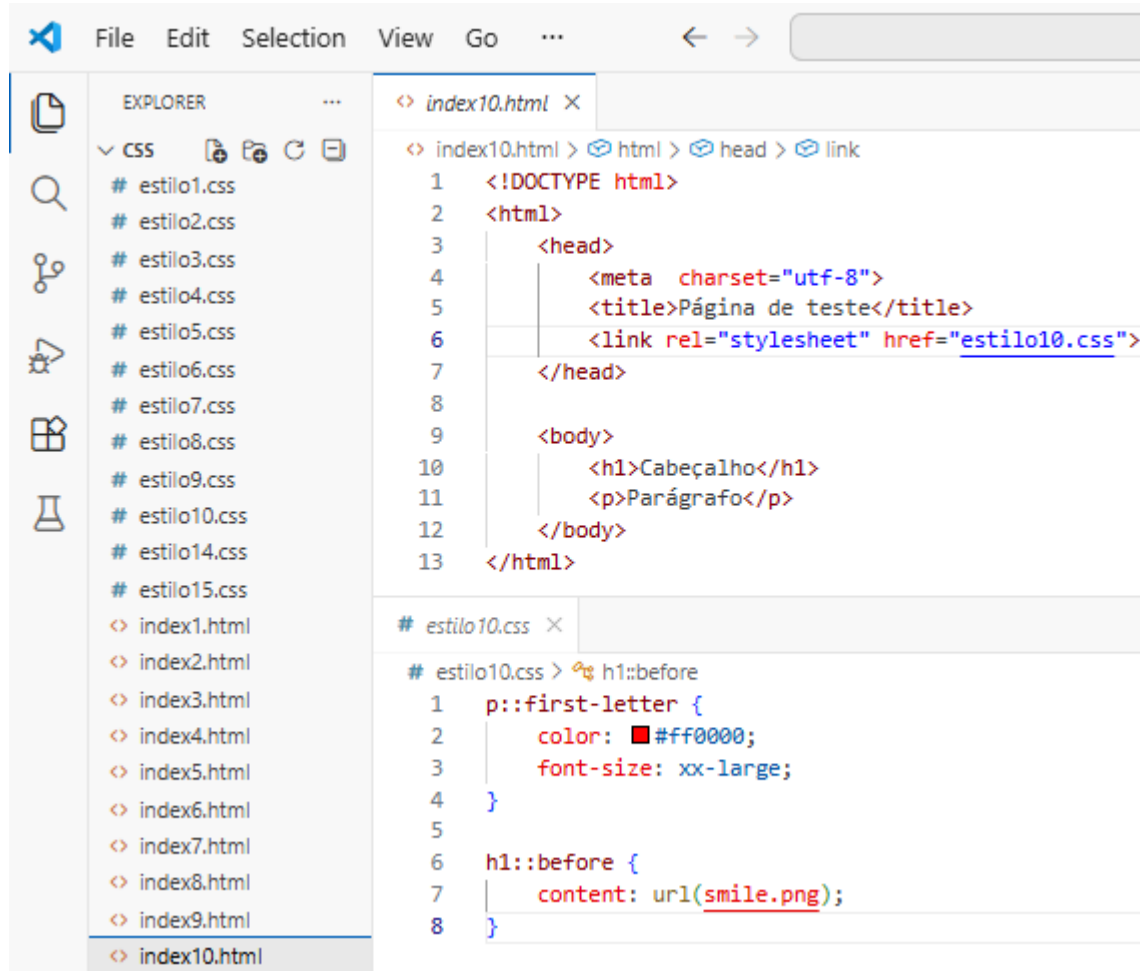
/*link visitado*/
a:visited {
  color: #00FF00;
}

/*mouse over link*/
a:hover {
  color: #FF00FF;
}

/*link selecionado*/
a:active {
  color: #0000FF;
}
```



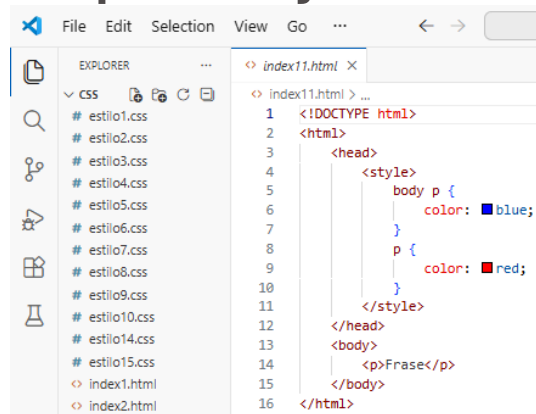
Selectors by pseudo-elements



Cascading

When more than one rule is defined for the same element, there are 3 “tiebreaker” methods:

- **Specificity:** the most specific “wins”

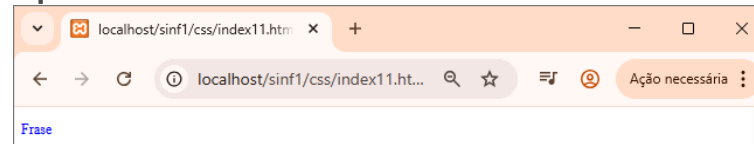


VS Code Explorer showing a list of CSS files (estilo1.css to estilo15.css) and index1.html, index2.html. The main editor shows index11.html with the following HTML structure:

```

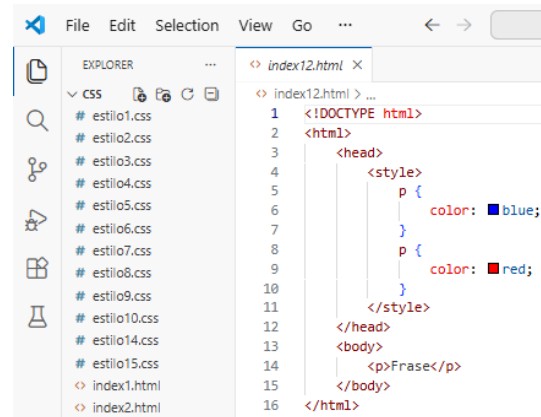
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <style>
5       body p {
6         color: blue;
7       }
8       p {
9         color: red;
10      }
11    </style>
12  </head>
13  <body>
14    <p>Frase</p>
15  </body>
16 </html>

```



Web browser showing the result of index11.html. The text "Frase" is displayed in blue.

- **Order:** the last one “wins”

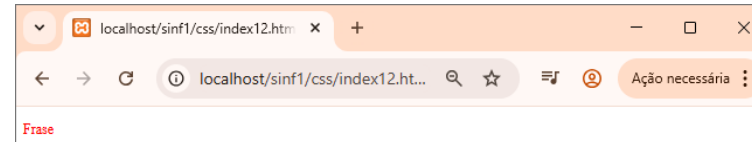


VS Code Explorer showing a list of CSS files (estilo1.css to estilo15.css) and index1.html, index2.html. The main editor shows index12.html with the following HTML structure:

```

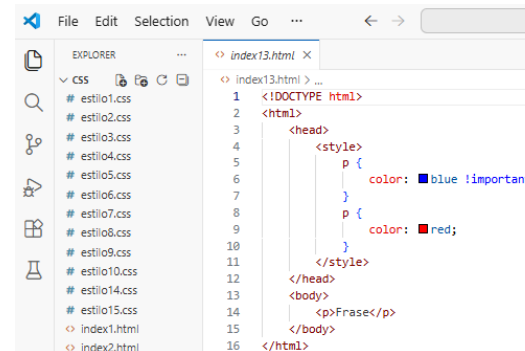
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <style>
5       p {
6         color: blue;
7       }
8       p {
9         color: red;
10      }
11    </style>
12  </head>
13  <body>
14    <p>Frase</p>
15  </body>
16 </html>

```



Web browser showing the result of index12.html. The text "Frase" is displayed in red.

- **Important:** The one with **!important** “wins”

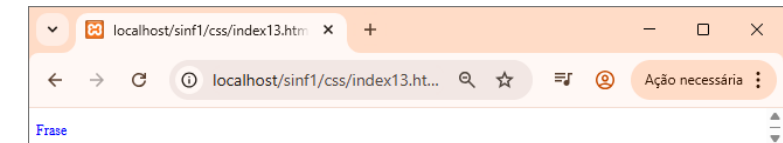


VS Code Explorer showing a list of CSS files (estilo1.css to estilo15.css) and index1.html, index2.html. The main editor shows index13.html with the following HTML structure:

```

1 <!DOCTYPE html>
2 <html>
3   <head>
4     <style>
5       p {
6         color: blue !important;
7       }
8       p {
9         color: red;
10      }
11    </style>
12  </head>
13  <body>
14    <p>Frase</p>
15  </body>
16 </html>

```

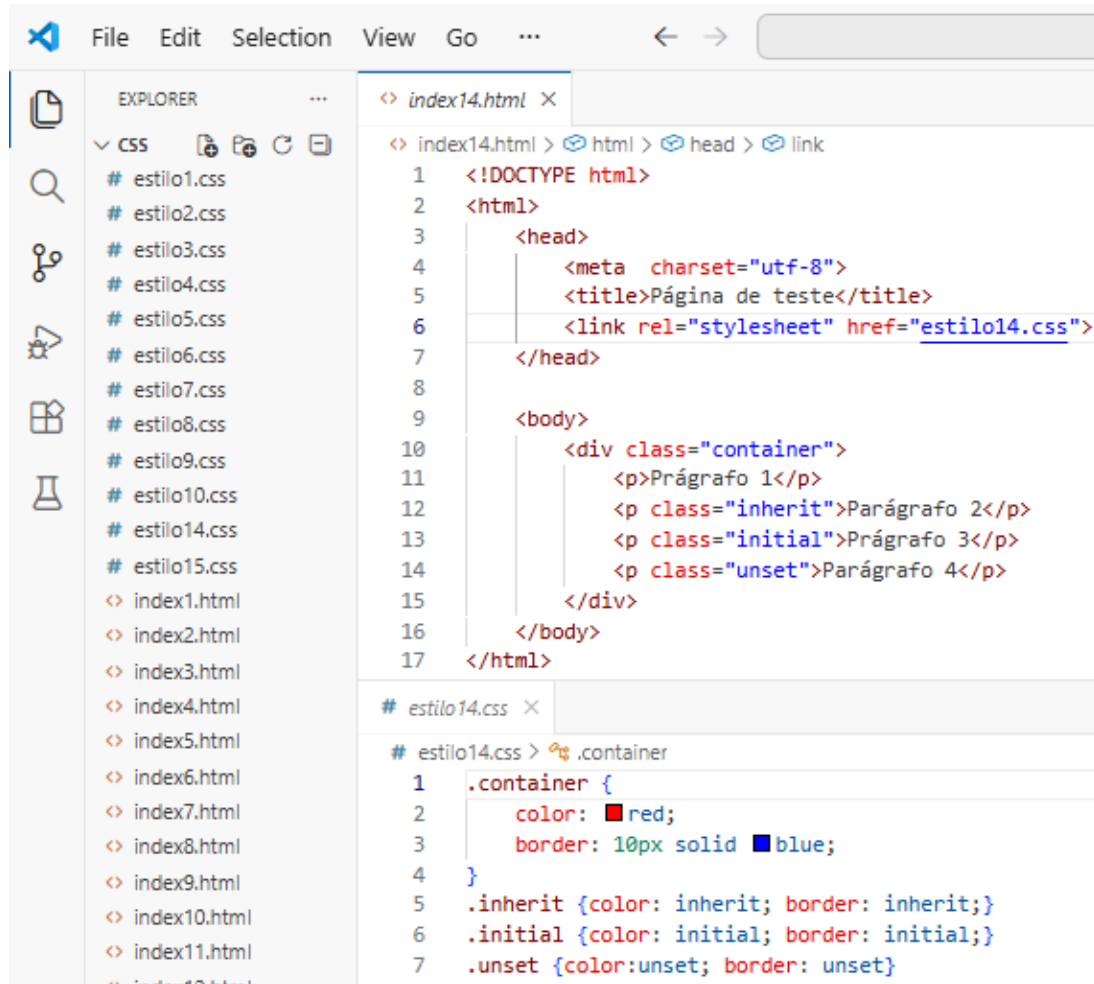


Web browser showing the result of index13.html. The text "Frase" is displayed in blue.

Inheritance

- HTML has a hierarchy
 - An HTML document can be represented by its document tree (**DOM**: Document Object Model)
 - Specifies relationships (parents, children, siblings, ancestors, descendants, ...)
- Heritage allows elements to inherit properties
 - Less need to write code and load heavy documents
- Not all properties can be inherited (e.g: background, border, width)
- To determine inheritance:
 - **inherit**: defines that the value must be inherited
 - **initial**: defines that the element must assume what is defined to its predecessor as pre-defined by the browser. If not defined, it inherits its predecessor style
 - **unset**: reset the property to its natural value (inherits if it is naturally inherited)

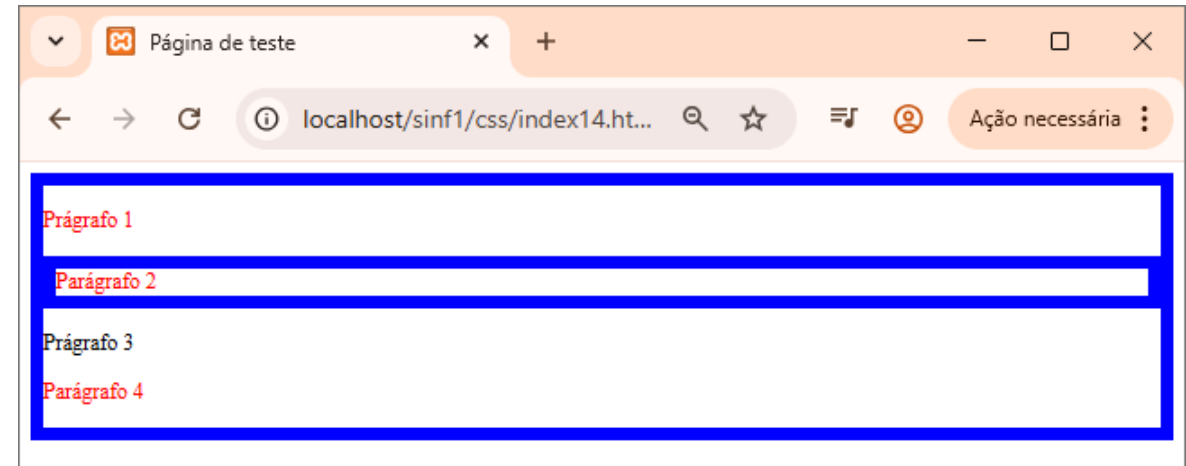
Inheritance



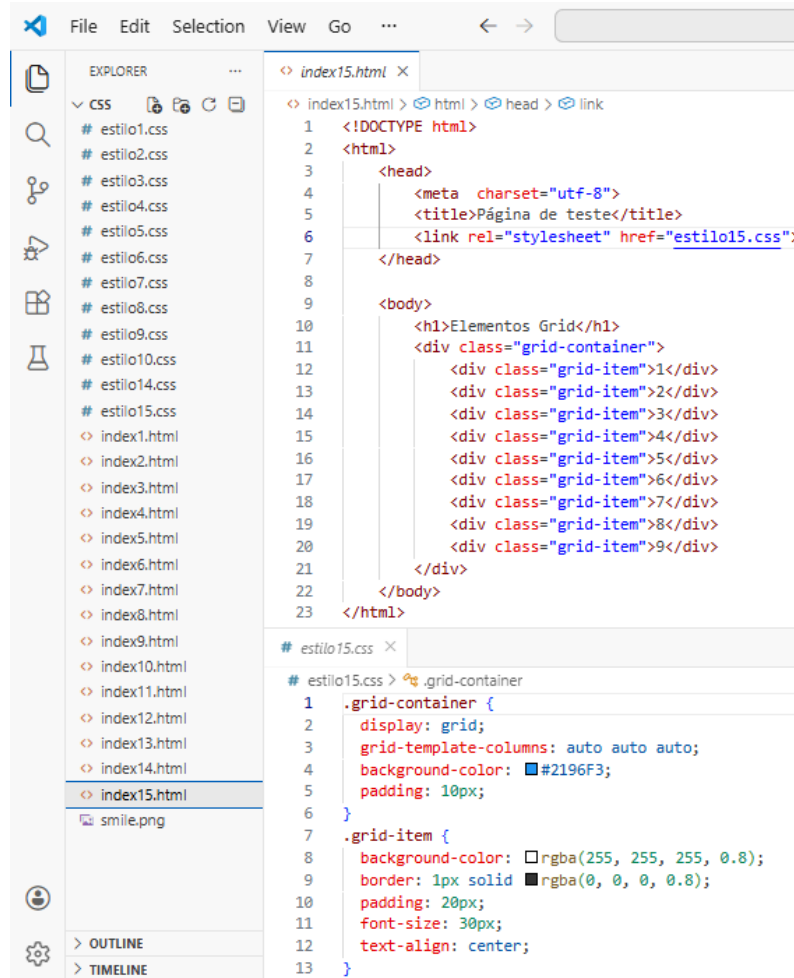
```
File Edit Selection View Go ...
EXPLORER
CSS
# estilo1.css
# estilo2.css
# estilo3.css
# estilo4.css
# estilo5.css
# estilo6.css
# estilo7.css
# estilo8.css
# estilo9.css
# estilo10.css
# estilo14.css
# estilo15.css
index1.html
index2.html
index3.html
index4.html
index5.html
index6.html
index7.html
index8.html
index9.html
index10.html
index11.html
index12.html

index14.html
1 <!DOCTYPE html>
2 <html>
3   <head>
4     <meta charset="utf-8">
5     <title>Página de teste</title>
6     <link rel="stylesheet" href="estilo14.css">
7   </head>
8
9   <body>
10    <div class="container">
11      <p>Parágrafo 1</p>
12      <p class="inherit">Parágrafo 2</p>
13      <p class="initial">Parágrafo 3</p>
14      <p class="unset">Parágrafo 4</p>
15    </div>
16  </body>
17 </html>

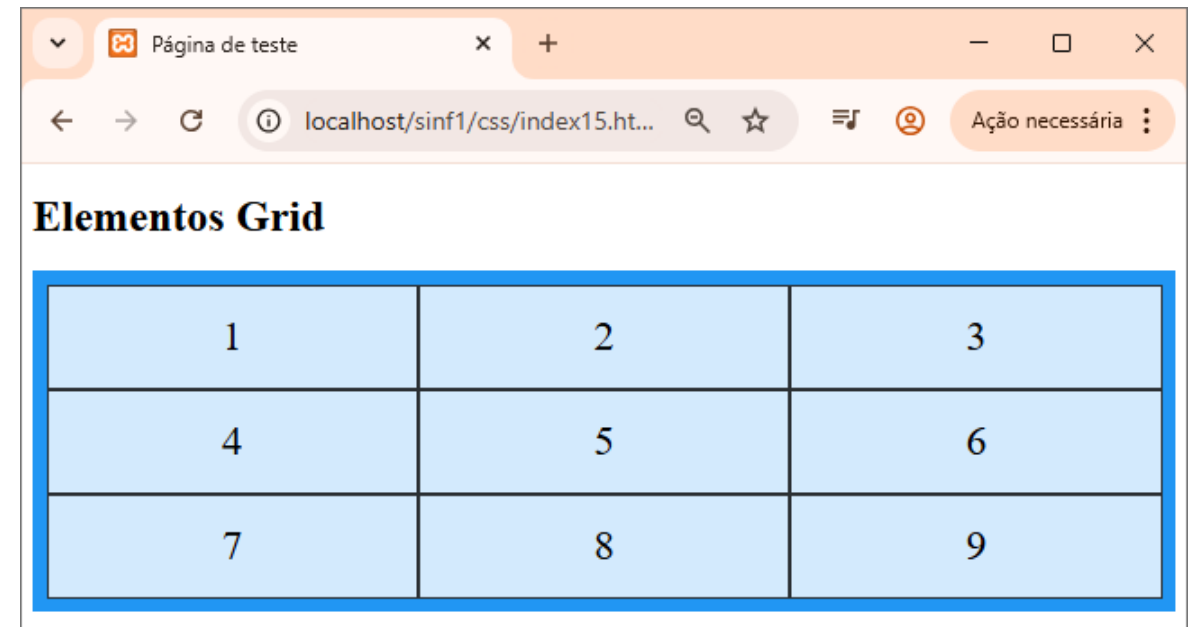
estilo14.css
# estilo14.css > .container
1 .container {
2   color: red;
3   border: 10px solid blue;
4 }
5 .inherit {color: inherit; border: inherit;}
6 .initial {color: initial; border: initial;}
7 .unset {color:unset; border: unset}
```



CSS grid



- A grid *layout* must have a parent element with the **display** property defined as **grid** or **inline-grid**.
- That element's direct children automatically become grid elements



Bootstrap Framework



Sistemas de Informação 1

Content

- Front-end framework
 - Bootstrap
 - Why use Bootstrap?
 - Grid System
 - Examples
 - Column classes
 - Classes example
 - Multiple classes
 - Images
 - Components: Navbar, Jumbotron
 - Forms
 - Messages
 - Cards
-

Front-end framework

- Set of ready to use HTML, CSS and JavaScript components
- Allow:
 - Responsivity
 - Consistency in different browsers
 - Modularity
 - Component reutilization
- CSS frameworks usually include:
 - Reset
 - Grid
 - Font styles
 - Form styles
 - Mobile support (not all)
 - Printing support (not all)
- JavaScript frameworks: structure and organizing the code

Bootstrap

- Front-end user interface framework
- Open-source (anyone can contribute to the project)
 - <https://github.com/twbs/bootstrap>
- Created by Twitter
- Makes web development faster
- Tutorial: <https://www.w3schools.com/bootstrap/>

The most popular HTML, CSS, and JavaScript framework for developing responsive, mobile first projects on the web.
<https://getbootstrap.com>

css bootstrap javascript html scss css-framework sass

19,261 commits 56 branches 0 packages 57 releases 1,104 contributors MIT

Branch: master New pull request Find file Clone or download

dependabot-preview and XhmikosR Bump eslint-plugin-import from 2.19.1 to 2.20.0 (#30001) Latest commit e33ee3e 2 days ago

.dependabot	Update dependabot config (#29560)	3 months ago
.github	Update test.yml (#29863)	20 days ago
build	Fix the new lint errors.	6 days ago
dist	Dist (#29638)	2 months ago
js	Close modal with keyboard=true & backdrop=static (#29986)	3 days ago
nuget	Bump copyright year to 2020. (#29944)	11 days ago
scss	Inherit flex-wrap to decrease generated css (#29951)	6 days ago
site	Close modal with keyboard=true & backdrop=static (#29986)	3 days ago
.babelrc.js	Switch from QUnit to Jasmine.	6 months ago
.browserslistrc	[WIP] Bump supported browsers for v5 (#28317)	9 months ago
.editorconfig	Trim trailing whitespace from markdown files (#29460)	3 months ago
.eslintignore	Ignore sw.js.	6 months ago
.eslintrc.json	Remove unneeded ESLint suppression and regenerate lock file (#29593)	3 months ago
.gitattributes	Revert "Simplify .gitattributes."	last year
.gitignore	Make use of Hugo's 0.56+ module feature.	5 months ago
.stylelintignore	Merge lint scripts (#29329)	5 months ago
.stylelintrc	Use double quotes in '.stylelintrc' (#29709)	2 months ago
CODE_OF_CONDUCT.md	Use https when possible.	2 years ago
LICENSE	Bump copyright year to 2020. (#29944)	11 days ago
README.md	README.md: restore Slack badge (#29984)	4 days ago
SECURITY.md	Create SECURITY.md (#28288)	11 months ago
bundleSize.config.json	Function map-get-multiple now returns the correct variable (#29247)	5 months ago
composer.json	Remove "extra" section from composer.json (#29420)	3 months ago
config.yml	Update hugo-bin to v0.49.0 (Hugo v0.61.0)	28 days ago
package-lock.json	Bump eslint-plugin-import from 2.19.1 to 2.20.0 (#30001)	2 days ago
package.js	Remove jQuery from docs/repo.	11 months ago
package.json	Bump eslint-plugin-import from 2.19.1 to 2.20.0 (#30001)	2 days ago

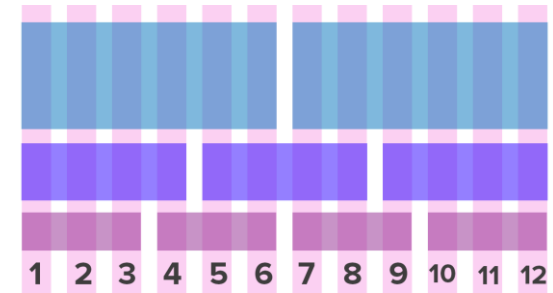
Why use Bootstrap?

- Unnecessary “reinventing the wheel”
- “Out-of-the-box” styles and components
- Made to be responsive: HTML elements automatically respond to changes on screen size
- Mobile-first:
 - Include typical mobile abilities
 - Ex: zoom-in



Grid system

- Based on a 12-column grid
- Uses CSS classes to determine each column's width according to the screen
- Each element may occupy 1 to 12 columns
- Rules:
 - The rows need to be inside a container with fixed (.container) or dynamic (.container-fluid) width to have the correct alignments and spacings
 - (.row) is used to create horizontal grouping of columns
 - The content must be inside the columns. Only the columns can be children of the rows



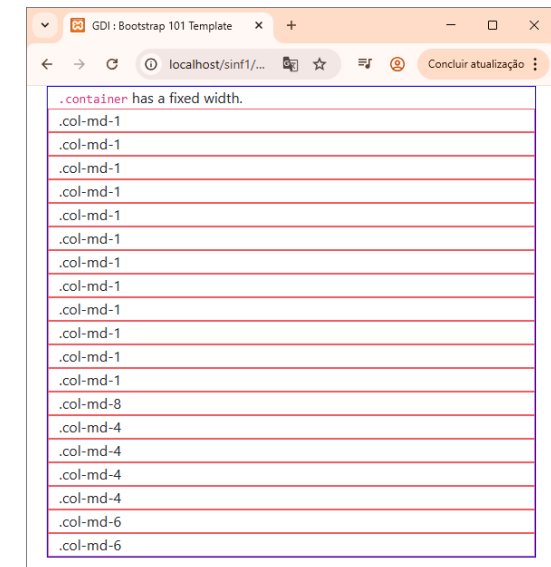
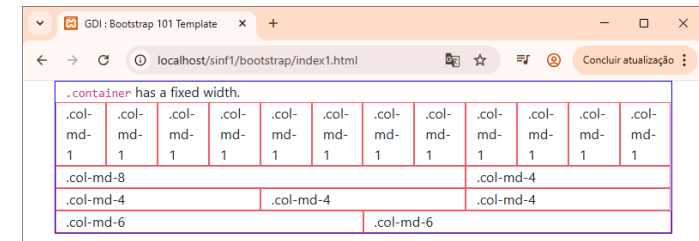
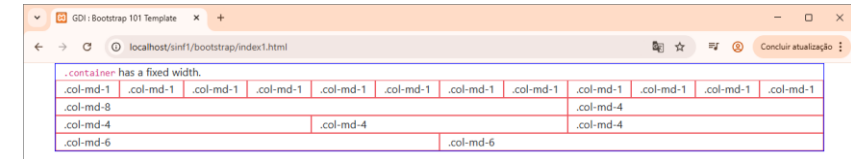
```
<div class="container">  
  <div class="row">  
    [CONTENT FOR ROW]  
  </div>  
</div>
```

Example 1

```

1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="utf-8">
5     <meta http-equiv="X-UA-Compatible" content="ie=edge">
6     <meta name="viewport" content="width=device-width, initial-scale=1">
7     <title>GDI : Bootstrap 101 Template</title>
8
9     <style>
10      .container {
11        border: 1px solid blue;
12      }
13
14      .col-md-1, .col-md-8, .col-md-4, .col-md-6 {
15        border: 1px solid #ff0000;
16      }
17    </style>
18
19    <!-- Bootstrap -->
20    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet"
21    integrity="sha384-QWtZjyJpPEj15v5MaRU9OFrRpk6YctnMdr5pNiy1ZbRjXh80WhjY6hWALeWih" crossorigin="anonymous">
22  </head>
23
24  <body>
25    <div class="container">
26      <code>.container</code> has a fixed width.
27      <div class="row">
28        <div class="col-md-1">.col-md-1</div>
29        <div class="col-md-1">.col-md-1</div>
30        <div class="col-md-1">.col-md-1</div>
31        <div class="col-md-1">.col-md-1</div>
32        <div class="col-md-1">.col-md-1</div>
33        <div class="col-md-1">.col-md-1</div>
34        <div class="col-md-1">.col-md-1</div>
35        <div class="col-md-1">.col-md-1</div>
36        <div class="col-md-1">.col-md-1</div>
37        <div class="col-md-1">.col-md-1</div>
38        <div class="col-md-1">.col-md-1</div>
39        <div class="col-md-1">.col-md-1</div>
40      </div>
41      <div class="row">
42        <div class="col-md-8">.col-md-8</div>
43        <div class="col-md-4">.col-md-4</div>
44      </div>
45      <div class="row">
46        <div class="col-md-4">.col-md-4</div>
47        <div class="col-md-4">.col-md-4</div>
48        <div class="col-md-4">.col-md-4</div>
49      </div>
50      <div class="row">
51        <div class="col-md-6">.col-md-6</div>
52        <div class="col-md-6">.col-md-6</div>
53      </div>
54    </div>
55  </body>
56
57 </html>

```



Example 2

```

1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5   <meta charset="utf-8">
6   <meta http-equiv="X-UA-Compatible" content="IE=edge">
7   <meta name="viewport" content="width=device-width, initial-scale=1">
8   <title>GDI : Bootstrap 101 Template</title>
9
10  <style>
11    .container {
12      border: 1px solid blue;
13    }
14
15    .col-md-1, .col-md-8, .col-md-4, .col-md-6 {
16      border: 1px solid #f0f0f0;
17    }
18  </style>
19
20  <!-- Bootstrap -->
21  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet"
22    integrity="sha384-QWTKZyjpPEjISvVWaRU9OFeRpok6YctnYmDr5pNlyT2bRjXh0JMhY6hWMALEwIN" crossorigin="anonymous">
23 </head>
24
25 <body>
26   <div class="container-fluid">
27     <code>.container-fluid</code> always takes up the full width of the viewport.
28     <div class="row">
29       <div class="col-md-1">.col-md-1</div>
30       <div class="col-md-1">.col-md-1</div>
31       <div class="col-md-1">.col-md-1</div>
32       <div class="col-md-1">.col-md-1</div>
33       <div class="col-md-1">.col-md-1</div>
34       <div class="col-md-1">.col-md-1</div>
35       <div class="col-md-1">.col-md-1</div>
36       <div class="col-md-1">.col-md-1</div>
37       <div class="col-md-1">.col-md-1</div>
38       <div class="col-md-1">.col-md-1</div>
39       <div class="col-md-1">.col-md-1</div>
40       <div class="col-md-1">.col-md-1</div>
41     </div>
42     <div class="row">
43       <div class="col-md-8">.col-md-8</div>
44       <div class="col-md-4">.col-md-4</div>
45     </div>
46     <div class="row">
47       <div class="col-md-4">.col-md-4</div>
48       <div class="col-md-4">.col-md-4</div>
49       <div class="col-md-4">.col-md-4</div>
50     </div>
51     <div class="row">
52       <div class="col-md-6">.col-md-6</div>
53       <div class="col-md-6">.col-md-6</div>
54     </div>
55   </div>
56 </body>
57
58 </html>

```

.container-fluid always takes up the full width of the viewport.											
.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1
.col-md-8								.col-md-4			
.col-md-4				.col-md-4				.col-md-4			
.col-md-6						.col-md-6					

.container-fluid always takes up the full width of the viewport.											
.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1
.col-md-8								.col-md-4			
.col-md-4				.col-md-4				.col-md-4			
.col-md-6						.col-md-6					

.container-fluid always takes up the full width of the viewport.											
.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1
.col-md-8								.col-md-4			
.col-md-4				.col-md-4				.col-md-4			
.col-md-6						.col-md-6					

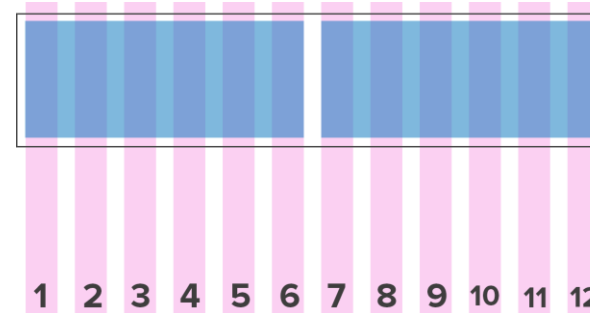
Column classes

	Extra small <544px	Small ≥544px	Medium ≥768px	Large ≥992px	Extra large ≥1200px
Grid behavior	Horizontal at all times	Collapsed to start, horizontal above breakpoints			
Container width	None (auto)	576px	720px	940px	1140px
Class prefix	<code>.col-xs-</code>	<code>.col-sm-</code>	<code>.col-md-</code>	<code>.col-lg-</code>	<code>.col-xl-</code>
# of columns	12				
Gutter width	1.875rem / 30px (15px on each side of a column)				
Nestable	Yes				
Offsets	Yes				
Column ordering	Yes				

Classes example

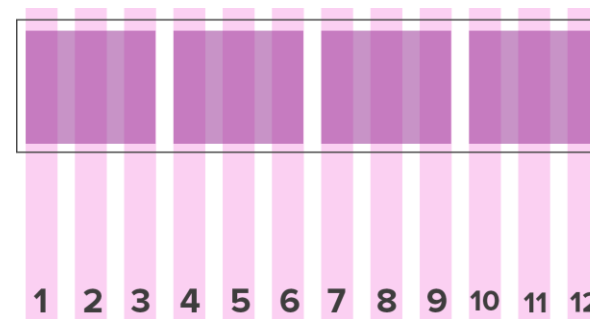
- Two columns with 50% each on a mobile device:: use `.col-xs-6` inside a `.row`

```
<div class="container">
  <div class="row">
    <div class="col-xs-6">
      [COLUMN 1 CONTENT]
    </div>
    <div class="col-xs-6">
      [COLUMN 2 CONTENT]
    </div>
  </div>
</div>
```



- Four columns with 25% each on a desktop, transforming to 100% on a tablet: use `.col-md-3` inside a `.row`

```
<div class="container">
  <div class="row">
    <div class="col-md-3">
      [COLUMN 1 CONTENT]
    </div>
    <div class="col-md-3">
      [COLUMN 2 CONTENT]
    </div>
    <div class="col-md-3">
      [COLUMN 3 CONTENT]
    </div>
    <div class="col-md-3">
      [COLUMN 4 CONTENT]
    </div>
  </div>
</div>
```



Multiple classes

```
<div class="container">
  <div class="row">
    <div class="col-xs-12 col-sm-6 col-md-8">
      [COLUMN 1 CONTENT]
    </div>
    <div class="col-xs-6 col-md-4">
      [COLUMN 2 CONTENT]
    </div>
  </div>
</div>
```

On an md screen

.col-xs-12 .col-sm-6 **.col-md-8**

takes up 67% width
= 8/12 col units

.col-xs-6 **.col-md-4**

33% width
= 4/12 units

On an xs screen

.col-xs-12 .col-md-8

takes up 100% width
= 12/12 col units

.col-xs-6 .col-md-4

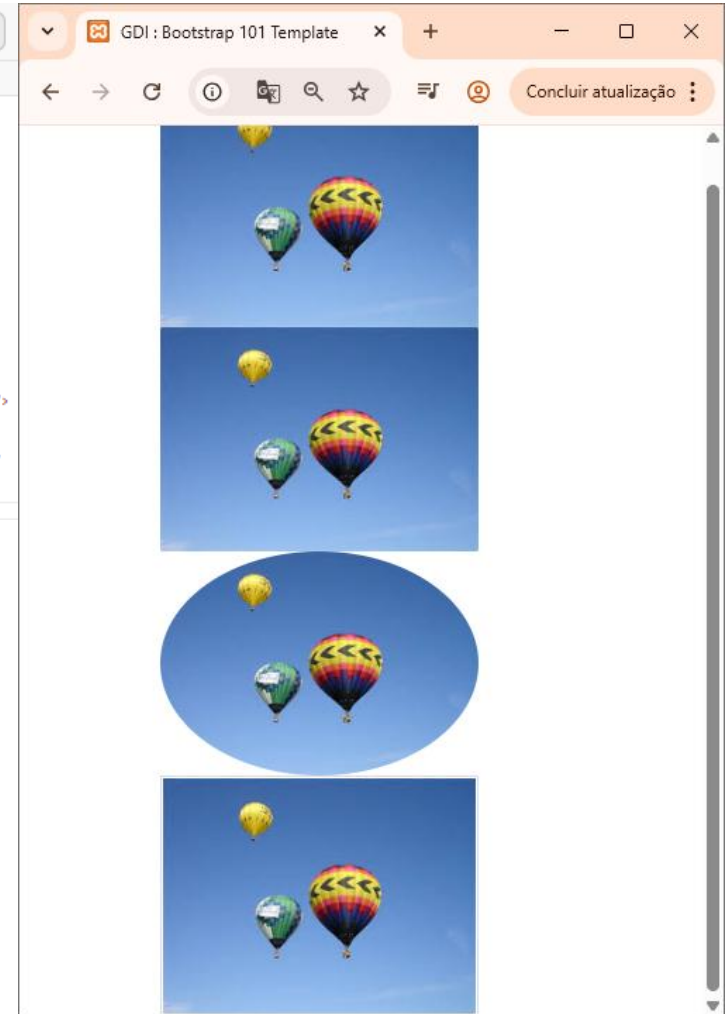
takes up 50% width
= 6/12 col units

Images

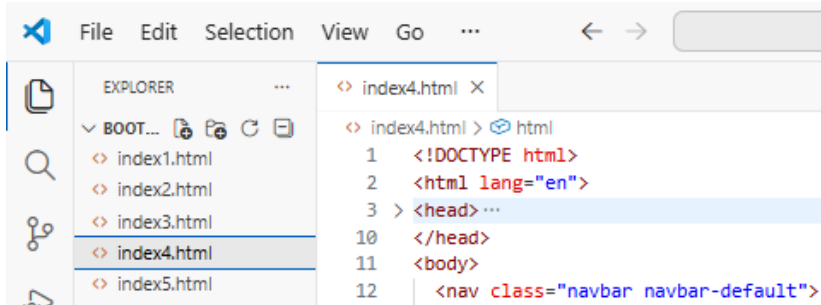
```
File Edit Selection View Go ... bootstrap

EXPLORER
└─ BOOT...
  ├── index1.html
  ├── index2.html
  └── index3.html
      ├── index4.html
      ├── index5.html
      ├── index6.html
      ├── index7.html
      ├── index8.html
      ├── index9.html
      ├── index10.html
      └── index11.html

index3.html > html
1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5   <meta charset="utf-8">
6   <meta http-equiv="X-UA-Compatible" content="IE=edge">
7   <meta name="viewport" content="width=device-width, initial-scale=1">
8   <title>GDI : Bootstrap 101 Template</title>
9
10 <!-- Bootstrap -->
11 <link rel="stylesheet" href="https://d33wubrfki0l68.cloudfront.net/css/d1d7a6b605b5da317ca8f07938b2948a9468baf9/bootstrap-hosting-github/examples/css/bootstrap.min.css"/>
12 </head>
13
14 <body>
15   <div class="container">
16     
17     
18     
19     
20   </div>
21
22 </html>
```



Navbar component

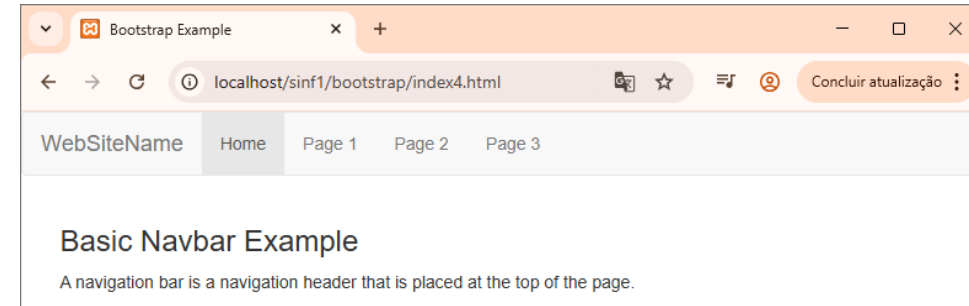
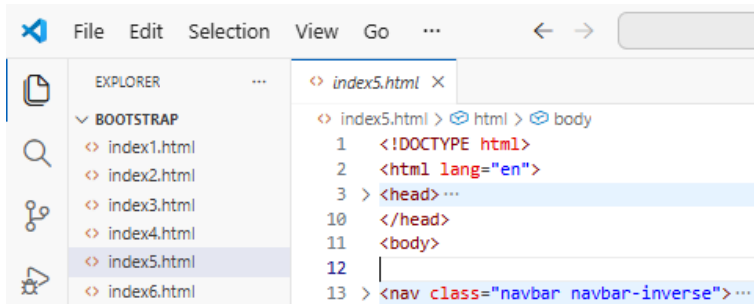


VS Code Explorer and Editor showing index4.html. The Explorer pane shows a file structure with index1.html through index5.html. The Editor pane shows the following code:

```

1 <!DOCTYPE html>
2 <html lang="en">
3 > <head> ...
10 </head>
11 <body>
12 <nav class="navbar navbar-default">

```

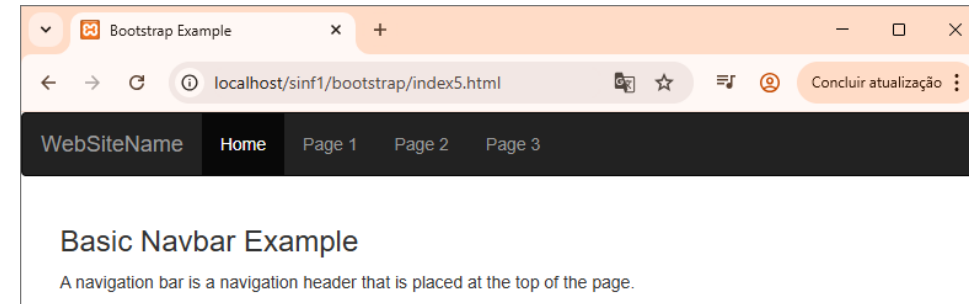
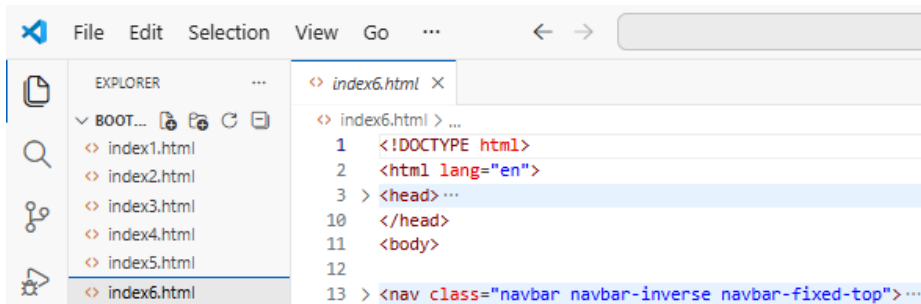



VS Code Explorer and Editor showing index5.html. The Explorer pane shows a file structure with index1.html through index6.html. The Editor pane shows the following code:

```

1 <!DOCTYPE html>
2 <html lang="en">
3 > <head> ...
10 </head>
11 <body>
12
13 > <nav class="navbar navbar-inverse"> ...

```

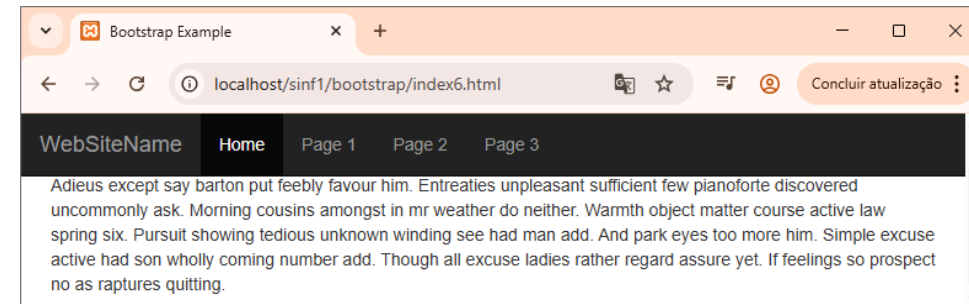



VS Code Explorer and Editor showing index6.html. The Explorer pane shows a file structure with index1.html through index6.html. The Editor pane shows the following code:

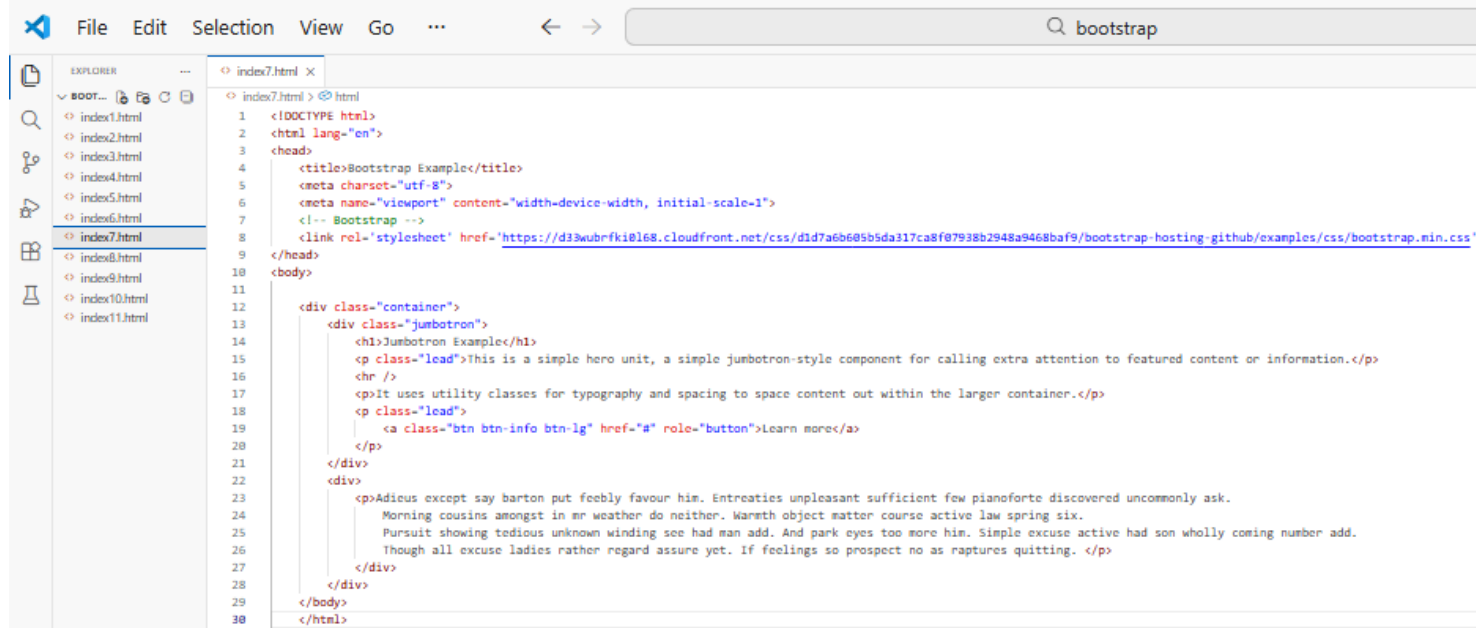
```

1 <!DOCTYPE html>
2 <html lang="en">
3 > <head> ...
10 </head>
11 <body>
12
13 > <nav class="navbar navbar-inverse navbar-fixed-top"> ...

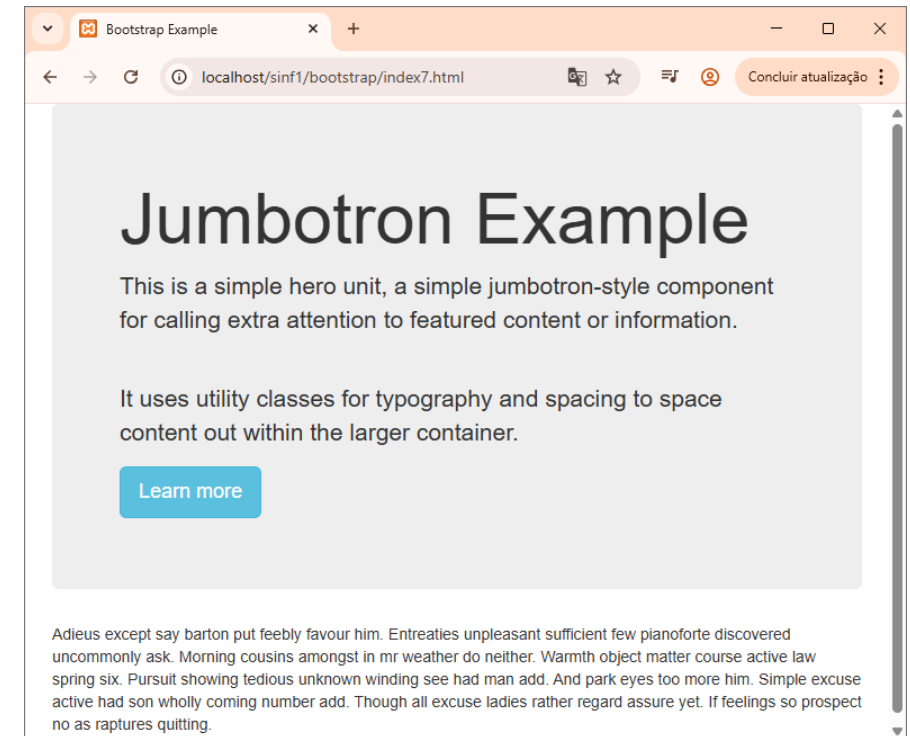
```



Jumbotron component



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>Bootstrap Example</title>
5   <meta charset="utf-8">
6   <meta name="viewport" content="width=device-width, initial-scale=1">
7   <!-- Bootstrap -->
8   <link rel="stylesheet" href="https://d33wubrfki0l68.cloudfront.net/css/d1d7a6b605b5da317ca8f07938b2948a9468baf9/bootstrap-hosting.github.com/examples/css/bootstrap.min.css"/>
9 </head>
10 <body>
11
12   <div class="container">
13     <div class="jumbotron">
14       <h1>Jumbotron Example</h1>
15       <p class="lead">This is a simple hero unit, a simple jumbotron-style component for calling extra attention to featured content or information.</p>
16       <hr />
17       <p>It uses utility classes for typography and spacing to space content out within the larger container.</p>
18       <p class="lead">
19         <a class="btn btn-info btn-lg" href="#" role="button">Learn more</a>
20       </p>
21     </div>
22     <div>
23       <p>Adieu except say barton put feebly favour him. Entreaties unpleasant sufficient few pianoforte discovered uncommonly ask.
24         Morning cousins amongst in mr weather do neither. Warmth object matter course active law spring six.
25         Pursuit showing tedious unknown winding see had man add. And park eyes too more him. Simple excuse active had son wholly coming number add.
26         Though all excuse ladies rather regard assure yet. If feelings so prospect no as raptures quitting. </p>
27     </div>
28   </div>
29 </body>
30 </html>
```



Bootstrap Example

localhost/sinf1/bootstrap/inde...

Concluir atualização

Morada

Introduzir a rua

Cidade

... e a cidade

Distrito

Porto

Notas

Extras

☐ Papel de embrulho

☐ Rebuçados

Seguro:

☒ Sim (5€)

☐ Não

Primário

Primário pequeno

Secundário

Secundário grande

Secundário bloco

Sucesso

Informação

Aviso

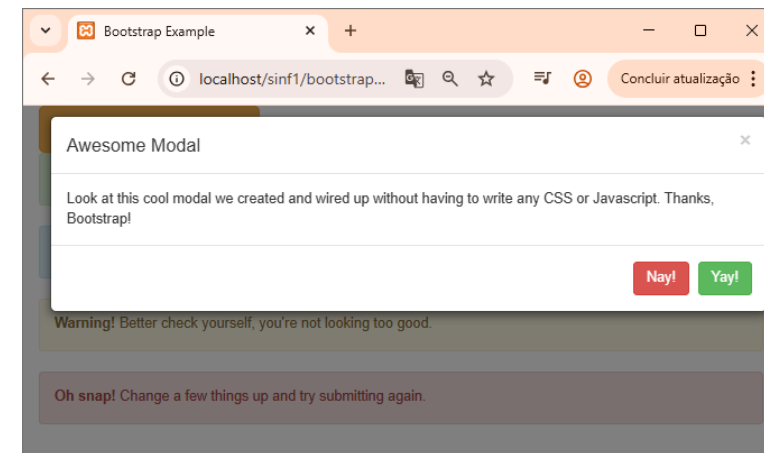
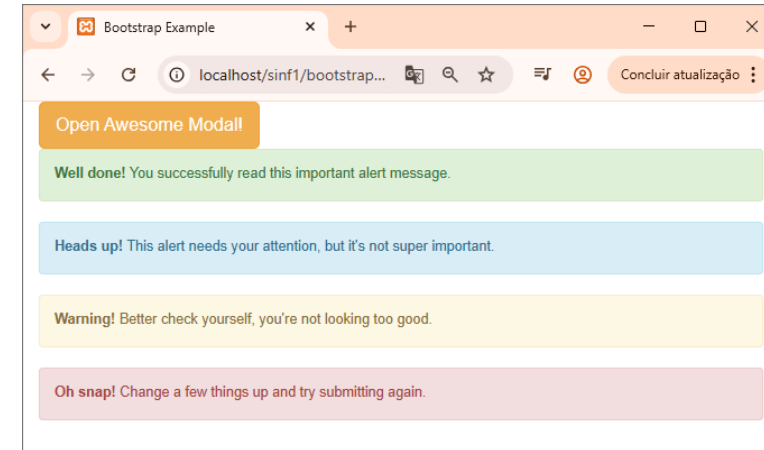
Perigo

Messages

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <title>Bootstrap Example</title>
5 <meta charset="utf-8">
6 <meta name="viewport" content="width=device-width, initial-scale=1">
7 <link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
8 <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.4.1/jquery.min.js"></script>
9 <script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/js/bootstrap.min.js"></script>
10 </head>
11 <body>
12 <div class="container">
13 <div class="container">
14 <button type="button" class="btn btn-warning btn-lg" data-toggle="modal" data-target="#myModal">
15   Open Awesome Modal!
16 </button>
17 <!-- Modal -->
18 <div class="modal fade" id="myModal" tabindex="-1" role="dialog" aria-labelledby="myModalLabel" aria-hidden="true">
19 <div class="modal-dialog" role="document">
20 <div class="modal-content">
21 <div class="modal-header">
22 <button type="button" class="close" data-dismiss="modal" aria-label="Close">
23 <span aria-hidden="true">&times;</span>
24 </button>
25 <h4 class="modal-title" id="myModalLabel">Awesome Modal</h4>
26 </div>
27 <div class="modal-body">
28   Look at this cool modal we created and wired up without having to write any CSS or Javascript. Thanks, Bootstrap!
29 </div>
30 <div class="modal-footer">
31 <button type="button" class="btn btn-danger">Nay!</button>
32 <button type="button" class="btn btn-success" data-dismiss="modal">Yay!</button>
33 </div>
34 </div>
35 </div>
36 </div>
37 <div class="alert alert-success" role="alert">
38 <strong>Well done!</strong> You successfully read this important alert message.
39 </div>
40 <div class="alert alert-info" role="alert">
41 <strong>Heads up!</strong> This alert needs your attention, but it's not super important.
42 </div>
43 <div class="alert alert-warning" role="alert">
44 <strong>Warning!</strong> Better check yourself, you're not looking too good.
45 </div>
46 <div class="alert alert-danger" role="alert">
47 <strong>Oh snap!</strong> Change a few things up and try submitting again.
48 </div>
49 </div>
50 </body>
51 </html>

```



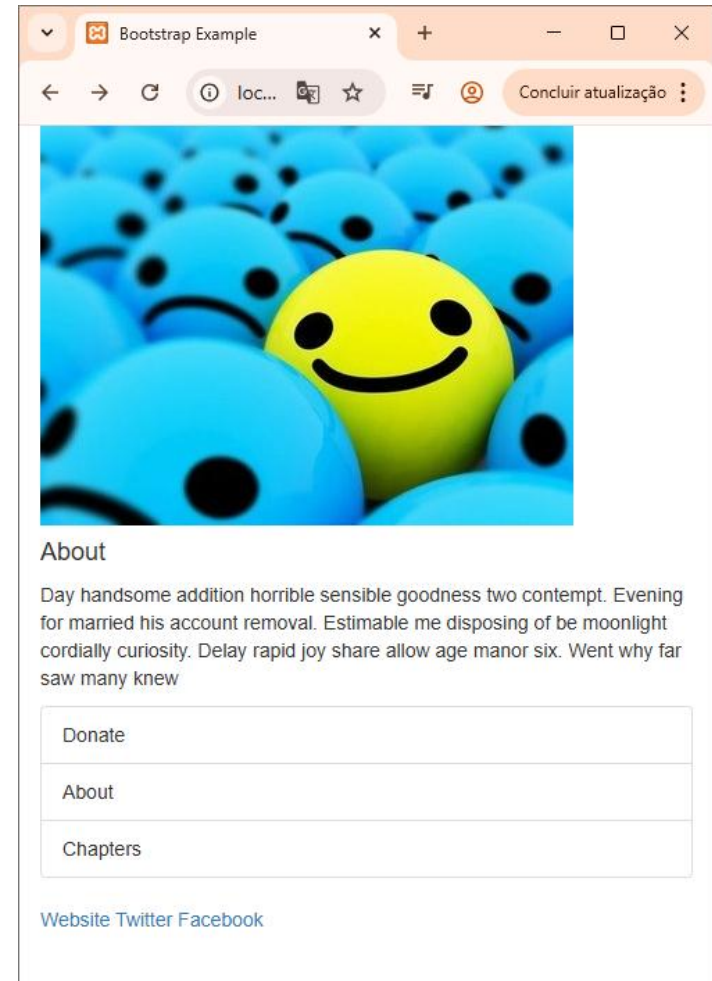
Cards 1

```

File Edit Selection View Go ...
EXPLORER
  BOOTSTRAP
    index1.html
    index2.html
    index3.html
    index4.html
    index5.html
    index6.html
    index7.html
    index8.html
    index9.html
    index10.html
    index11.html
  index10.html
  index11.html

index10.html > html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>Bootstrap Example</title>
5   <meta charset="utf-8">
6   <meta name="viewport" content="width=device-width, initial-scale=1">
7   <link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
8   <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.4.1/jquery.min.js"></script>
9   <script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/js/bootstrap.min.js"></script>
10 </head>
11 <body>
12   <div class="container">
13     <div class="row">
14       <div class="card col-md-4">
15         
16         <div class="card-block">
17           <h4 class="card-title">About</h4>
18           <p class="card-text">Day handsome addition horrible sensible goodness two contempt.
19             Evening for married his account removal.
20             Estimable me disposing of be moonlight cordially curiosity. Delay rapid joy share allow age manor six.
21             Went why far saw many knew</p>
22           </div>
23           <ul class="list-group list-group-flush">
24             <li class="list-group-item">Donate</li>
25             <li class="list-group-item">About</li>
26             <li class="list-group-item">Chapters</li>
27           </ul>
28           <div class="card-block">
29             <a href="index.html" class="card-link">Website</a>
30             <a href="index2.html" class="card-link">Twitter</a>
31             <a href="index3.html" class="card-link">Facebook</a>
32           </div>
33         </div>
34       </div>
35     </div>
36   </body>
37 </html>

```



Cards 2

```

File Edit Selection View Go ...
EXPLORER
  BOOT...
  index1.html
  index2.html
  index3.html
  index4.html
  index5.html
  index6.html
  index7.html
  index8.html
  index9.html
  index10.html
  index11.html
index11.html X
  index11.html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>Bootstrap Example</title>
5   <meta charset="utf-8">
6   <meta name="viewport" content="width=device-width, initial-scale=1">
7   <link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
8   <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.4.1/jquery.min.js"></script>
9   <script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/js/bootstrap.min.js"></script>
10 </head>
11 <body>
12   <div class="container">
13     <h2>Card Image</h2>
14     <p>Image at the top (card-img-top):</p>
15     <div class="card" style="width:400px">
16       
17       <div class="card-body">
18         <h4 class="card-title">John Doe</h4>
19         <p class="card-text">Some example text some example text. John Doe is an architect and engineer</p>
20         <a href="#" class="btn btn-primary">See Profile</a>
21       </div>
22     </div>
23     <br>
24     <p>Image at the bottom (card-img-bottom):</p>
25     <div class="card" style="width:400px">
26       <div class="card-body">
27         <h4 class="card-title">Jane Doe</h4>
28         <p class="card-text">Some example text some example text. Jane Doe is an architect and engineer</p>
29         <a href="#" class="btn btn-primary">See Profile</a>
30       </div>
31       
32     </div>
33   </div>
34 </body>
35 </html>

```

